Inner-Outer Point Evaluation in a Deterministic Generalization Domain

Pablo S. Mandolesi†, Osvaldo Agamennoni‡,
Guillermo Sentoni†and Alfredo Desages‡
Departamento de Ingeniería Eléctrica
Universidad Nacional del Sur
Av. Alem 1253, (8000), Bahía Blanca, Argentina
†C.O.N.I.C.E.T.
‡C.I.C.
e-mail: iemandol@criba.edu.ar

August 24, 2004

Abstract

In this paper the generalization domain of a given model is addressed. The generalization domain is usually related with the polytope formed by the training data. Three different approaches to evaluate if a point belongs to a given n-dimensional convex polytope are presented. The first strategy reduce the dimensionality of the problem. The second one make uses of some geometrical consideration to apply Linear programming technique as well as Linear Matrix Inequality methods. The last approach is focused on the simplification of the polytope by removing interior points or producing a simple new polytope in which the original one is included.

Keywords**:** Generalization Domain, Polytope, Validation.

1 INTRODUCTION

All mathematical models are built to reproduce the behavior of a system under certain conditions. A model may be identified through the adjustment of its parameters using a given finite set of input-output data. It is desirable to achieve a good model performance not only for the training data but also for any other input belonging to a given input domain. In this respect, we have two fundamental problems to study: the generalization property, i.e. how well a model reproduces the behaviour of a real system, and the domain or region of the input space where the model shows thsi good behaviour, i.e. the generalization domain. It is not easy to measure the generalization property of a model. Attemps to deal with this problem can be found in Weeb [20] and Holden et. al. [7] and the references therein. However, the measurement of the generalization property is difficult and usually restricted to binary inputs. The evaluation of the generalization properties and the generalization domain of the model are two key problems in many areas such as control, prediction etc.. A real time measure that indicates if a model input is inside or outside the generalization domain would be an important contribution. The present paper is focused on the evaluation of a real time indicator or flag to know if a model is driven inside or otside the generalization domain.

There exist some statistical approaches to study the generalization properties of a Neural Network model. An introductory theoretical framework may be found in [23]. Basically, a generalization measure allows to estimate how large a training set is needed for good generalization. Good means that the probable error is smaller than some convenient number. The domain of generalization is a related different approach. This is a "model" that describes the model applicability, i.e. the domain in the input space were good generalization is expected. No consideration of the network or model architecture is made.

There is no precise definition of the generalization domain. However, it is usually related with the polytope of the training domain, or better, the region delimited by the convex hull covering the training set. On the other hand, if we perform a model test to validate its performance, it is possible to identify a cluster in the input space were that performace is acceptable. Beside a quantification of that performace may be evaluated, and in all cases a fast evaluation if a given model input belong to a given polytope or not is important. In other words, we have to evaluate if a point ($\mathbf{x}_c$) is inside or

outside a given polytope. This problem, has many other different application in mathematics, computer graphics, etc.

To associate the interpolation domain with the generalization domain we need to assume that:

- Good approximation without overfitting is achieved.
- All data in the training shell are uniformly distributed, i.e. the density of points per volume unit is nearly constant.

There exists well established algorithm to meet the first assumption. It is not always possible to meet the second assumption. If a non convex training shell is covered with a convex one, this will have zones without any data. In this case a clustering algorithm may be used to divide the non convex shell in a set of connected convex shells.

Let us assume that m points in $\Re^n$, form a polytope $\Im$. The evaluation if a point belongs to a given polytope has a very simple solution for $n \leq 2$. A Neural Network solution of the problem with $n = 2$ can be found in Jayadeva and Bhahumik [8]. However, the complexity of the problem grows exponentially with n. Courrier [4] presented a method based on calculating the distance between $\mathbf{x}_c$ and $\Im$. If this distance is zero, $\mathbf{x}_c$ belongs to $\Im$; otherwise $\mathbf{x}_c$ is outside. Courrieu also presented an approximation of this method by using recurrent neural networks and another simple one by using the polytope's circumscribed sphere.

In this work we present three different approaches. The first one is a finite algorithm approach. We develop a mechanism to reduce the problem of determining if the origin belongs to $\Im$, to the same problem in a reduced order space polytope $\Im^{n-1}$. Then successive applications of the same method can be used to solve the problem. The second approach makes use of an optimization algorithm. This approach improves the optimization based algorithm of Courrieu [4]. With a simple consideration, a linear programming algorithm may be used to solve the problem [9] based on its reformulation taking into account some geometric considerations. A connection of this problem with the linear matrix inequalities [3] problem is also provided.Finally the third approach is based on the covering of the polytope by a hyperrectangle. The hyperrectangle faces are evaluated with the principal component vectors. In this way to check if a point is inside the domain or not we only need to evaluate a matrix product. In this case, the presence of isotropically distributed

clusters and/or outliers with a high noise level introduce a wrong enlargement in the generalization domain. There are some techniques to circumvent this problem. @@faltan las referencias de esto ultimo

Finally, three different examples are presented to show the potentials of the present results on modelling the applicability or the generalization domain. The first is a nonlinear static functions showing the differences on the behaviour of a feedforward neural network between the generalization domain and the extrapolation domain. The last two examples illustrate the generalization domain evaluation problem of a dynamic model. Dynamic modellling introduces a new aspect to be consider: the input memory. We will concentrate our discussion in systems with finite memory and we will show that it is not always possible to define the generalization domain from the input domain. An approach to solve this problem is also presented..

2 Problem Statement

Let

$$\mathbf{x} = (x_1, x_2, \cdots, x_n)^T, \mathbf{x} \in \Re^n \tag{1}$$

and let

$$X = \{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m\}$$

form a polytope in $\Re^n$ by its elements convex combination, i.e.,

$$\Im = \Im(X) = \{\mathbf{x} | \, \mathbf{x} = \sum_{\mathbf{x}_i \in X} \lambda_i \, \mathbf{x}_i, \sum_{i=1 \cdots m} \lambda_i = 1, \lambda_i \geq 0\}. \tag{2}$$

Let $\mathbf{x}_c$ be any point in $\Re^n$

$$\mathbf{x}_c = (x_{c1}, x_{c2}, \cdots, x_{cn})^T, \mathbf{x}_c \in .$$

We want to determine if $\mathbf{x}_c$ is inside or outside the polytope defined by (2). There is no loss of generality in assuming $\mathbf{x}_c$ in the origin. Then X can be redefined as follows

$$X = \{\mathbf{x}_1 - \mathbf{x}_c, \mathbf{x}_2 - \mathbf{x}_c, \cdots, \mathbf{x}_m - \mathbf{x}_c\}.$$

From now on we will consider $\mathbf{x}_c$ to be the origen.

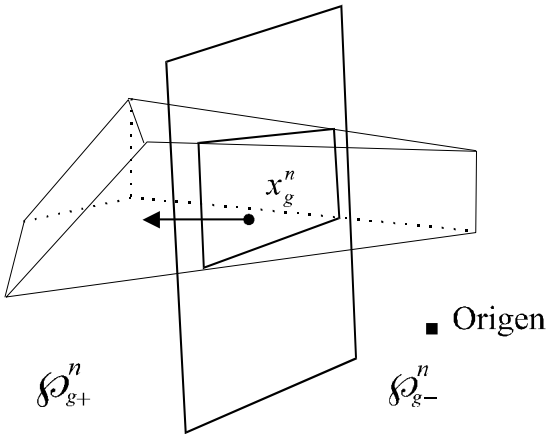


Figure 1: Dimension reduction by hyperplane separation

2.1 Dimension Reduction Case

In this subsection, a superscrip to note the dimension of the space will be added to clarify the presentation of the algorithm ($\Im^n = \Im \in R^n$, $\mathbf{x}^n = \mathbf{x} \in R^n$, etc) since the algorithm that will be presented is based on a reduccion of the space demension. Let

$$\mathbf{x}_g^n = \frac{1}{m}\left(\sum_{i=1}^{m} \mathbf{x}_i^n\right)$$

the gravity center of the polytope $\Im^n$. It is clear that $\mathbf{x}_g^n$ belongs to $\Im^n$ since it is convex. Now take the hyperplane $\wp_g^n \in R^n$ defined as follows

$$\wp_g^n = \left\{ \mathbf{x}^n \,\middle|\, \left\langle \mathbf{x}^n, \mathbf{x}_g^n \right\rangle = 0 \right\}$$

which contains the origin and it is normal to $\mathbf{x}_g^n$. Then $\wp_g^n$ defines the two regions $\wp_{g+}^n$ and $\wp_{g-}^n$ of R^n, the positive and negative regions by the sides of $\wp_g^n$ respectively.

Now define $\Im^{n-1}(X^{n-1}) \in \wp_g^n$, $X^{n-1} = \{x_1^{n-1}, x_2^{n-1}, ...\}$, as the reduced order polytope, where $\mathbf{x}_i^{n-1} \in R^{n-1}$ are its vertix elements. The points $\mathbf{x}_i^{n-1}$ are the intersection points between the hiperplane $\wp^n$ and all the possible convex combination made with one point $\mathbf{x}_i^n \in \wp_{g+}^n$ and one $x_j^n \in \wp_{g-}^n$ of X^n. We would refer to them as the projection points and also $\Im^{n-1}$ as the *projection polytope* or the *reduced polytope*. Now it remains to show that if

the origen belongs to $\Im^n$, it also belongs to $\Im^{n-1}$. This is easily seen from the fact that $\wp_g^n$ contains the origen and the polytope is $\Im^{n-1}$ also convex.

Now we can sumarize this in the following algorithm:

1. Evaluate $\mathbf{x}_g^n$ to find $\wp_g^n$, group the points $\mathbf{x}_i^n$ in those who belong to $\wp_{g+}^n$ and $\wp_{g-}^n$ correspondingly, i.e.,

$$X_{g+}^n = X^n \cap \wp_{g+}^n = \left\{\mathbf{x}^n : \mathbf{x}_i^n \in X^n, \left\langle \mathbf{x}_i^n, \mathbf{x}_g^n \right\rangle > 0\right\}$$

and

$$X_{g-}^n = X^n \cap \wp_{g-}^n = \left\{\mathbf{x}^n : \mathbf{x}_i^n \in X^n, \left\langle \mathbf{x}_i^n, \mathbf{x}_g^n \right\rangle < 0\right\}.$$

2. Check if X_{g+}^n or X_{g-}^n are empty set $\emptyset$. In that case the origin does not belong to the polytope finishing the search. Otherwise, we reduce the polytope $\Im^n$ to its projection $\Im^{n-1}$.

$$\begin{aligned} \widehat{X}^n \;=\; & \left\{\widehat{\mathbf{x}}^n = \mu \mathbf{x}_{g+}^n - (1-\mu)\mathbf{x}_{g-}^n, \right. \\ & \left. \mu \,\middle|\, \mu \mathbf{x}_{g+}^n - (1-\mu)\mathbf{x}_{g-}^n \in \wp_g^n, \quad \mathbf{x}_{g+}^n \in X_{g+}^n, \quad \mathbf{x}_{g-}^n \in X_{g-}^n \right\} \end{aligned}$$

Now project $\widehat{X}^n$ over the $\wp_g^n$, i.e.

$$X^{n-1} = \widehat{X}^n \,/ \wp_g^n$$

Now set $n = n - 1$ and in the case that $n > 1$ go to 1 else continue with step 3.

3. At this time we have $\Im^1$ that is a segment and it is very easy to check if this segment contains the origin.

End.

This algorithm could have combinatorial growth in the number of points of the reduced dimension polytopes, i.e., $N(X_+)N(X_-)$, where $N(.)$ is the number of points of the polytope $\Im$ in $(.)$. The worst case occurs when $\wp_g$ divides the m points of the original polytope in two sets of $\frac{m}{2}$ points each, then $\Im^{n-1}$ will have $\frac{m^2}{4}$ points. Some of them will be inner points, but we have to consider all of them. If also $\wp_g^{n-1}$ divides $\Im^{n-1}$ in two set of $\frac{m^2}{8}$ points

we will have a $\Im^{n-2}$ polytope with $\frac{m^4}{64}$ points. The hyperplane $\wp_g$ provides an heuristic solution in minimizing $N(\wp_{g-})N(\wp_{g+})$ since $\mathbf{x}_g$ is a vector pointing to the highest density zone of points. Despite this combinatorial growth, the approach is very simple to implement and could be very efficient for some applications ($n \leq 3$). The performance of the algorithm may be improved by normalizing the politope on the unitary hypersphere, i.e. dividing each point of X^n by its modulus.

2.2 Optimization Based Strategy

Now we will drop the superscrip n since no change on the dimention of the space are needed. Let's now consider the hyperplanes $\wp_{\mathbf{x}_i}$ defined by the normal vectors $\mathbf{x}_i \in X$ and containing the origen. These planes divide the $\Re^n$ space in regions $\wp_{\mathbf{x}_i+}$ and $\wp_{\mathbf{x}_i-}$,

$$\wp_{\mathbf{x}_i+} = \{x \,|\langle x, \mathbf{x}_i\rangle > 0\} \quad \forall \mathbf{x}_i \in X$$
$$\wp_{\mathbf{x}_i-} = \{x \,|\langle x, \mathbf{x}_i\rangle < 0\} \quad \forall \mathbf{x}_i \in X.$$

Now it can be easily verified that if the origen belongs to the polytope, then:

$$\bigcup_{:\mathbf{x}_i \in X} \wp_{x_i+} = \Re^n - \{0\}. \tag{3}$$

or equivalently,

$$\bigcap_{:\mathbf{x}_i \in X} \wp_{x_i+} = \emptyset. \tag{4}$$

(See Fig. 2 for $n = 2$).

Then, from (4) we need to check the consistency of the following system of linear inequalities.

$$\langle \mathbf{x}_i, \mathbf{x}\rangle < 0 \;\; \forall\, \mathbf{x}_i \in X. \tag{5}$$

If this system of inequalities is feasible then there is at least one point that belongs to $\bigcap_{\mathbf{x}_i \in X} \wp_{\mathbf{x}_i-}$ and so (4) does not hold and the origin is outside the polytope.

To find the existence of a solution of equation (5), let us introduce a dummy variable t and solve the following linear programming problem [9],[5] :

$$\begin{array}{l} \min\limits_{\mathbf{x},t} t \\ \text{subject to } \langle \mathbf{x}_i, \mathbf{x}\rangle + t \geq 0, \;\; i = 1, \ldots, m. \end{array} \tag{6}$$

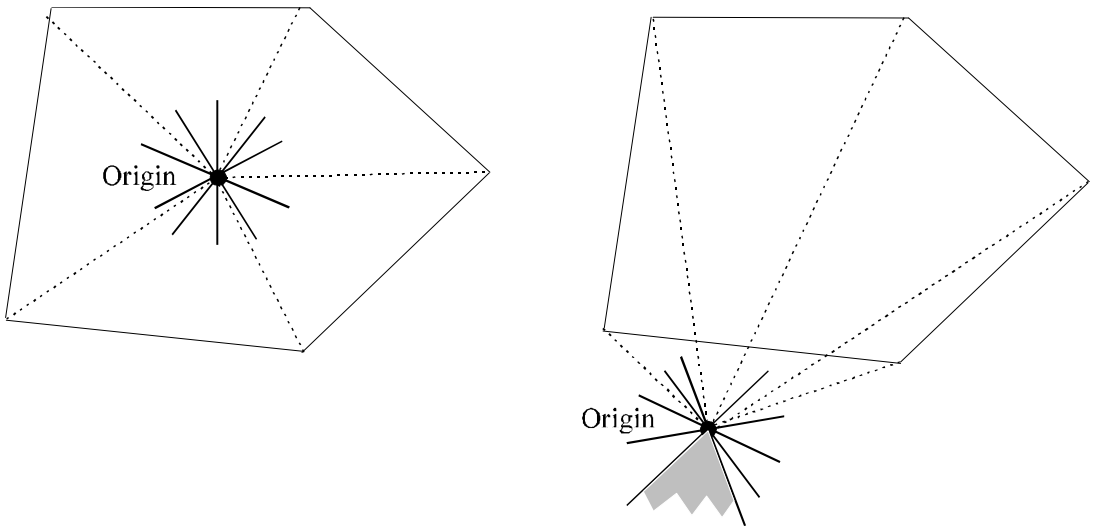


Figure 2: a) Inner point b) Outer point.

The sign of solution value t reveals the feasibility or not for the system of inequalities (5). If $t < 0$ there is no solution, so the polytope contains the origin, otherwise ($t > 0$) the origin is outside the polytope. In other words if $t \geq 0$ this means there exists no region in R^n where $\langle \mathbf{x}_i, \mathbf{x} \rangle$ can be simultaneously positive or simultaneously negative so there exists no solution to equation (5).

In many cases this minimization process can be unbounded, however the sign contains the information that we are looking for. In some cases in a standard linear programming program, an extra inequality can be added to the $\wp_{x_i+}$ constraints of (6) to prevent the unbounded cases. As an example [18] we have:

$$\left\langle \sum_{\mathbf{x}_i \in X} \mathbf{x}_i, \mathbf{x} \right\rangle \geq M,$$

where $M \in \Re$ is a big positive value.

This problem also can be solved using Linear Matrix Inequalities "LMI". The LMI problem [3] can be stated as follows: found the solution of the following inequality

$$\mathbf{F}_0 + \sum_{i=1,\ldots n} \mathbf{y}_i \, \mathbf{F}_i > 0, \tag{7}$$

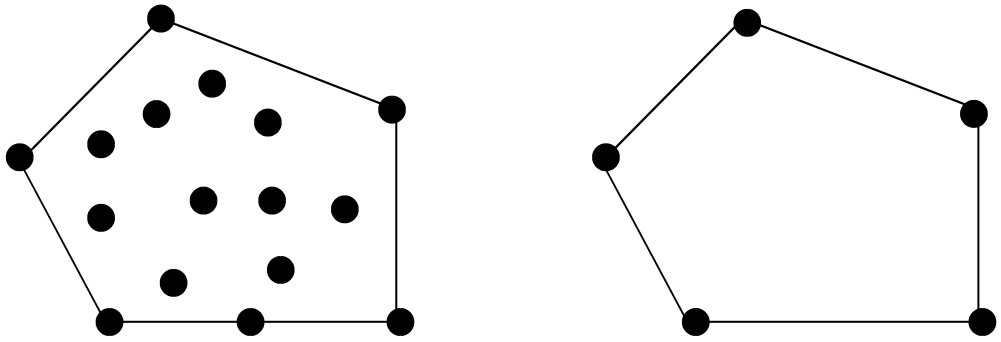


Figure 3: a) Generalization Domain b) Domain Shell

where $\mathbf{y} \in R^n$ is the variable and the symetric matrices $\mathbf{F}_i \in R^{nxn}$ are given.

Let $\mathbf{F}_i$ $i = 1, \ldots, n$, diagonal matrices of the form

$$\mathbf{F}_i = diag\{-x_{11}, -x_{21}, \ldots, -x_{j1}, \ldots\} \ \forall \mathbf{x}_j \in X$$

and $\mathbf{F}_0$ a null matrix with the same dimension as $\mathbf{F}_i$ be the coefficient matrix of the LMI problem. Now If problem (7) has a solution, then the origen will be outside the polytope.

The difference with respect to Courrieu's approach is that he solves a nonlinear optimization problem and we only need to check the feasibility of a given system of linear inequalities or LMI problem. Linear programming reduces complexity of the minimization process.

2.3 Problem Reduction

At the moment if a new point $\mathbf{x}$ is given, we can find whether or not it belongs to the generalization domain $\Im$. The size of the problem to be solved increases with the number of elements of X. The number of restrictions in the Linear Programing Problem (6) is the number of elements of X.

To simplify the problem two diferent approches can be proposed:

- Reduce the information available in X to the minumum needed. All the information we need are the exterior points. These points form a new polytope with the same volume and space positioning as the original one but with a minimum number of former points.

- On the other hand we can find an approximated domain establishing a new polytope with a special structure. This polytope is the one that covers the original domain and has a very simple structure. We will refer later to this problem as a domain subscription technique.

2.3.1 Interior Points Elimination

In this section we described an algorithm for reducing the size of the polytope which tends to be the shell of the generalization domain. We want to find a polytope $\tilde{\Im} = \tilde{\Im}\left(\tilde{X}\right)$ which expands the same volume as $\Im$ but with a smaller definition set

$$\tilde{X} = \{\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \cdots, \tilde{\mathbf{x}}_r\}.$$

So the number of points of $m > r$, where $m = cardinal(X)$ and $r = cardinal(\tilde{X})$. The smallest polytope $\tilde{\Im}$ is the shell of $\Im$, taking the minimum over the cardinal of the definition set. The polytope shell verify the following properties:

Property 1: The points $\tilde{\mathbf{x}}_i$ of the shell polytope verify

$$\tilde{\mathbf{x}}_i = \mathbf{x}_j \in X \mid \mathbf{x}_j \not\subset \Im(X - \mathbf{x}_j),$$

i.e., those points that belong to $\Im(X)$ and are outside the reduced polytope formed by the set $\tilde{\Im}(\{X - \mathbf{x}_j\})$.

Property 2: The shell points do not belong to any sub-polytope except those where they form part of its definition set.

Using these properties and selecting an iteration mechanism we form a simply interior point elimination algorithm. This algorithm can be described as follows:

1. Set

   $$X_0 = X$$

   the set to be reduced in the first iteration ($i = 0$). Then group in $\tilde{X}_0$ the first n linear independent points of X_0,

   $$\tilde{X}_0 = \{\mathbf{x}_{1,0}, \mathbf{x}_{2,0}, \cdots, \mathbf{x}_{n,0}\} \quad \mathbf{x}_{i,0} \quad i = 1, \cdots, n \quad L.I.,$$

   i.e., set $\tilde{\Im}_0$ that expands a volume in $\Re^n$.

2. Set $j = 1, m = cardinal(X_i)$

3. If $j \geq m$ then continue with step *4*. *Else,* check if $\mathbf{x}_{j,i}$ belongs to $\tilde{\Im}_i$, using any previous methods.
   If $\mathbf{x}_j$ is inside the partial generalizated domain defined by $\tilde{\Im}_i$, then $j = j + 1$ go to *3*. Else expand the generalization domain $\tilde{\Im}_i$ by adding $\mathbf{x}_j$ to it's definition set, $\tilde{X}_i = \tilde{X}_i \cup \{ \mathbf{x}_j \}$. Increment j, $j = j + 1$, and go to *3*.

4. if $m > cardinal(\tilde{X}_i)$ then
   Randomize the elements of $\tilde{X}_i$ and assign them to X_{i+1}, increment i, $i = i + 1$ and execute again from step *2*.
   Else continue with step *5*.

5. Now we have $\tilde{X}_i = \tilde{X}_{i-1}$ defining $\tilde{\Im}_i = \tilde{\Im}_{i-1}$ expanding the same volume as $\Im$.

The number of elements m of $\tilde{X}_i$ is close to the number of elements which form the shell, the $\tilde{\Im} \simeq \tilde{\Im}_i$, $(\tilde{X} \simeq \tilde{X}_i)$. The error in the number of points can be reduced by forcing the algorithm to continue running for a few times ignoring the break condition in step *4*.

2.3.2 Domain Subscription

Now we will introduce a different aproach to the generalization domain description. We subscribe the generalization domain in a hiper-parallelepiped and use this subscription for a fast evaluation on the domain inclusion.

Many formulations can be established to obtain the hiper-parallelepiped which covers the domain. To minimize the hiper-parallelepiped volume we use the structural information contained in the principal components. The eigenvertors related to each principal component are used as director vectors of the hiperplanes that form the faces. Each principal vector has two associated hyperplanes between which all data are contained. These two faces pass over the most distanced points in the director direction.

Let

$$\mathbf{x}_g = \frac{1}{m} \left(\sum_{i=1}^{m} \mathbf{x}_i \right)$$

be the center of mass of X^n, the vectors

$$\mathbf{b}_i^n = (b_{i1}, \cdots, b_{in})$$

are the eigenvectors of the principal components. Now we evaluate a signed distance of each point of the domain data set to the center of mass in the principal directions,

$$p_i = \langle \mathbf{b}_i, \mathbf{x}_j - \mathbf{x}_g \rangle \qquad \forall j = 1 \cdots m$$

then find the extreme distances, $p_{i\max}$ and $p_{i\min}$ to construct the two plane faces. In addition we choose the director vector of the plane pointing to the interior of the parallelepiped, making the domain evaluation more simple later. Taking into account these considerations the hiperplanes are:

$$\begin{aligned} \wp_{i\max} &= \{\mathbf{x} \,|\langle -\mathbf{b}_i, \mathbf{x}\rangle - \langle -\mathbf{b}_i, (\mathbf{x}_g + \mathbf{b}_i\, p_{i\max})\rangle = 0\}\,, \\ \wp_{i\min} &= \{\mathbf{x} \,|\langle \mathbf{b}_i, \mathbf{x}\rangle - \langle \mathbf{b}_i, (\mathbf{x}_g + \mathbf{b}_i\, p_{i\min})\rangle = 0\} \end{aligned}$$

as shown in Fig. **??**.

Now we will concentrate on the evaluation if a given new point belongs or not to the domain. We have two diferent ways:

1 At first, we may use the polytope formed by the parallelepiped vertices,

$$X = \{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m\} \quad m = 2^n$$

where $\mathbf{x}_i, i = 1, ..., m$ are the intersections of the $2n$ planes in groups on n excluding its paralell face. Then $\Im$ is the polytope with former set X. Now the inclusion can be evaluted by any of the expresed forms in the polytope evaluation strategy on the previus section.

2 Second,we may use the information contained in the faces hiperplanes. Let the selected direction vectors point to the interior of the hiperparalellpipe. Then we only need to evaluate the hiperplanes equation for the new point in consideration, if the results are all greater or equal to zero it belongs to the generalization domain. Otherwise, the point is outside. This evaluation can be done by a simple matrix multiplication,

$$s = \begin{bmatrix} -\mathbf{b}_1 & -\langle -\mathbf{b}_1, (\mathbf{x}_g + \mathbf{b}_1\, p_{1\max})\rangle \\ \mathbf{b}_1 & -\langle \mathbf{b}_1, (\mathbf{x}_g + \mathbf{b}_1\, p_{1\min})\rangle \\ & \\ \mathbf{b}_n & -\langle -\mathbf{b}_n, (\mathbf{x}_g + \mathbf{b}_n\, p_{n\max})\rangle \\ \mathbf{b}_n & -\langle \mathbf{b}_n, (\mathbf{x}_g + \mathbf{b}_n\, p_{n\min})\rangle \end{bmatrix} \cdot \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}.$$

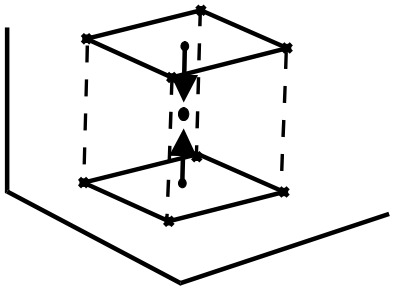


Figure 4: Two faces and its director vectors

Now if $\min s \geq 0$ it can be enssured that $\mathbf{x} \in \Im$ the generalization domain. Moreover, cheking if a point is inside or outside can be implemented very efficiently unsing Neural Network. Suppouse that we have a Single Hidden Layer Perceptron with: $n+1$ inputs, $(2n)$ hidden neurons, one output and hard limiting activation functions. Taking the equation of each hyperplanes as the weights and bias of each of the hidden neurons, and choosing conveniently the

3 Examples

In this section we will show three different examples to show the potentials on modelling the applicability or the generalization domain. The first is a nonlinear static function merely focused on showing the behaviour differences of a feedforward neural network in the generalization domain and in the extrapolation domain. In this first example the geometry of the generalization domain is easily defined from the input bounds. However, in the real world the shape of the generalization domian may be far from a simple geometrical figure.he last two example are focused on dynamic modelling. Dynamic modellling introduces a new aspect to be consider: the input memory. The memory of a system may be finite or infinite. We will concentrate our discussion in systems with finite memory . @@agregar referencias de modelacion pesados [2], [15] and [10]@@

If the input data set is long enought (teoretically infinite) and random with uniform distribution between two fixed values, then the generalization domain may be assumed as the inputs bounds box (see example 2). However, these assumptions are in general very difficult to achieve in practice. Even

thougt the input signals are bounded between fixed values the length of the sample data are not usually long enought to completelly cover all the input box. In other cases, the input signals shown some relationships or are constrained to a given particular region of the input space. In these cases the generalization domain can not be assumed as the input box.

A feedforward network may be used to reproduce the behaviour of a nonlinear finite memory dynamic system by using a tapped delay line of the input to obtain a FIR nonlinear model [15]. In this case, the dynamic information is translated to a static higher dimentional formulation. The generalization domain may be defined in the NN inputs space, i.e., the inputs of the system times the length of the tapped delay line. The dimentionality of a finite memory model may be very large, specially if the memory of the system is large. To reduce this dimentionality problem, different combinations of linear dynamics and nonlinear static models have been studied. The LNET structure is one of this attempts that combines a linear Laguerre dynamic system with a static perceptron [17]. Example 3 is dedicated to show how using this approach, the generalization domian definition is simplified.

3.1 Example 1:Nonlinear static function

Let assume the following function

$$y = \sum_{i=1}^{10} \sin(x_i)$$

For modelling this function, we trained a Multilayer Perceptron with two hidden layer of 10 neurons each by using 1000 randomly selected points in the hypercube $x_i \subset [0, 2\pi]$, $i = 1, 2, ..., 10$. We will asume this hypercube as the generalization domain. Figure 5 shows the square error in the approximation set, i.e. the set of samples used to train the network, the interpolation set, i.e. samples in the hypercube not used in the approximation and in the hypercube $x_i \subset [0, 2\pi]$, $i = 1, 2, ..., 9.$ and $x_{10} \subset [2\pi, 4\pi]$. It is important to note the change in the network performance. In this case the inner-outer evaluation evaluation if an input is inside or outside the generalization domain is direct due to its geometry. In practice, the generalization domain haven't a simple geometrical shape.

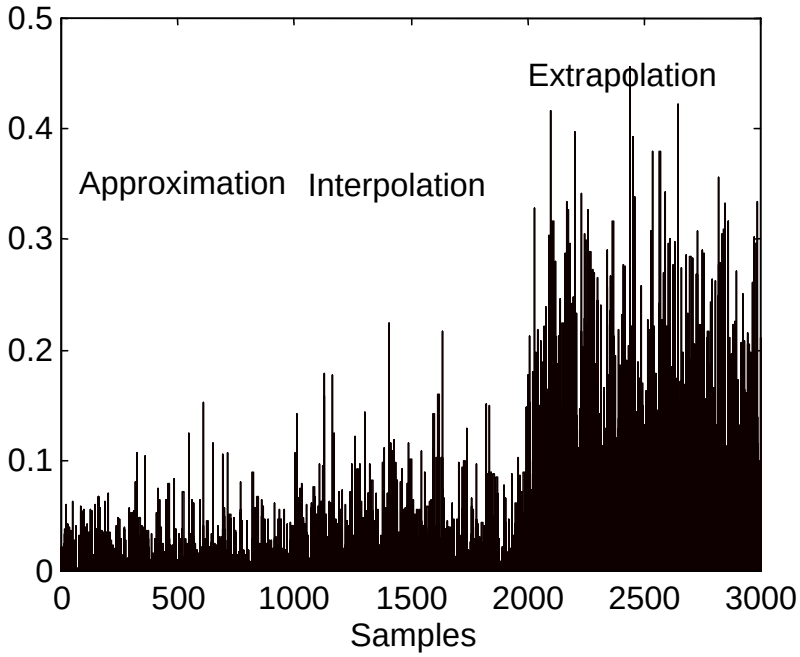


Figure 5: Square error

3.2 Example 2: Nonlinear dynamic system with a simple generalization domain

We will present a dynamic modelling with a simple generalization domain. In this case the generalization domain is directly evaluated from the input bounds. The nonlinear dynamic system

$$y\left(k+1\right)=\frac{y(k)}{1+y(k)^2}+\frac{0.5}{1+u_1(k)^2}+\frac{0.5}{1+u_2(k)^2}$$

has two decoupled inputs. We first generated a data set of four hunderd samples by using the following inputs

$$\begin{aligned} u_1 &= -2+0.01\,k \\ u_2 &= \text{randon with uniform dist. in } [-2,2] \end{aligned}$$

This kind of signals are usually present in systems in which it is not possible to manipulate all of their inputs. Biological systems (with the sustrate composition) or chemical systems (with the enviromental temperature) are examples of systems with unmanageable inputs. In the resent example, the input signals define a parallelogram in the plane u_1, u_2. We will assume this parallelogram a as the generalization domain. Figures **??** and **??** shows the inputs and the output of the system.

A line of ten tapped delay was consider in each input. Then we trained a feedforwad neural network with 2 hidden layers of 3 neurons each with the first half of samples data, i.e. with the first two hundred samples. The absolute error performance is plotted in Figure **??**. The mismatch between the network and the true system is small and uniform in the generalization domain. The absolute error in the extrapolation region shows an exponential increment with respecto the distance to the generalization domain. In this case the parallelogram formed by the input space represents an appropiate definition of the generalization domain. The behavior of the model is clearly different inside and outside the generalization domain. The next example shows that the generalization domain could not alwasy be defined from the input polytope.

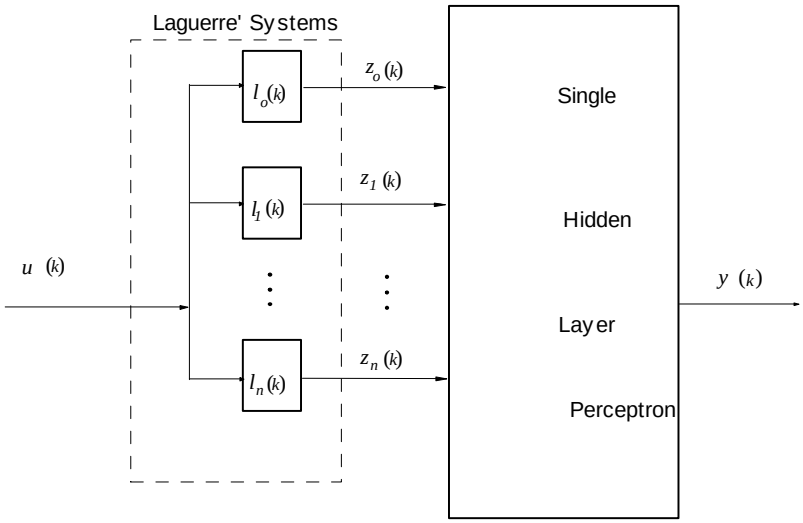


Figure 6: LNet architecture

3.3 Example 3: Nonlinear dynamical system with a non trivial generalization domain.

The present example is to show that not always is possible to evaluate a generalization domain from the input bounds. For this purpose we will use the LNet structure for approximating nonlinear discrete systems. This modelling approach was introduced by Sentoni et al. and a complete description of the technique may be found in [17]. This approach consist on a special interconextion of a linear Laguerre systen with a multilayer perceptron. The only necessary previous knowledge of the systems is reflected in choosing the Laguerre pole, closely related to its main time constant. In Fig.6, we can see the approximation structure composed by a set of discrete time Laguerre systems and a Single Hidden Layer Perceptron.

This model structure is especially suitable for validation by viewing the generalization domain. The linear part is completely defined by the pole, while the nonlinear part is only a static map. As can be seen from the Fig. 6, the input signal is filtered by the Laguerre systems:

$$L_i(z) = \frac{C}{z-a}\left[\frac{1-az}{z-a}\right]^i, i: 0, ..., mo \tag{8}$$

where: $L_i(z)$ is the z-transform of $l_i(k)$, the ith order discrete Laguerre System, $C = \sqrt{(1-a^2)\,T}$, T the sample time.The generating pole a is the inverse of the dominat time constant of the systems. These filters turn the one dimensional signal onto a multidimensional one, then mapped by the Single Hidden Layer Perceptron. The pressent example will illustrate those points.

Let the system be the one described by the following equations:

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k) \\ s(k) &= \mathbf{C}\mathbf{x}(k) + \mathbf{D}u(k) \\ y(k) &= s^3(k) - 6s^2(k) + 8.5s(k) + 1.5 \end{aligned}$$

where:

$$\mathbf{A} = \begin{bmatrix} .8 & 0 & 0 \\ .6 & .8 & .8 \\ .48 & -.36 & .8 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$
$$\mathbf{C} = \begin{bmatrix} .6 & .3 & .204 \end{bmatrix}, \mathbf{D} = [0].$$

This example was designed to test the behavior of the model inside and outside the generalization domain. For this purpouse, we have generated two data groups: *group 1* (into the generalization domain, with $u(k) \in [-.4, .1]$), and *group 2* (into the generalization and extrapolation domains, with $u(k) \in [-.1, .5]$). The range of the *group 1* corresponds enterely to the α *section* of the system's nonlinearity, while range of the *group 2* belongs jointly to the α *and* β *sections* (Fig. **??**). The data *group 1* was random splitted into a training set α_{tr} and a test set α_{ts}.

The LNet that approximates this system was chosen to be composed by three Laguerre systems ($mo = 3$) with the pole $a = 0.8$, and a single hidden Layer Perceptron with seven neurons. *The LNet model was adjusted using only the training data set* α_{tr}. The performance of this model can be seen in Fig. 8 for both the training and the test sets α_{tr} and α_{ts} respectively. The matching is very good, and the differences can only be detected by the error plot. It must be emphasized that this test was realized in the generalization domain. Now we will use the data *group 2* to test the generalization into the entire domain of the α *and* β *sections*. In the Fig. 9 can be appreciated three plots: a) the system and LNet outputs; b) the logarithmic of the absolute approximation error; and c) the inside-outside indicator flag. The behavior of the model for the time running from 100 to 225 is very poor, as can be

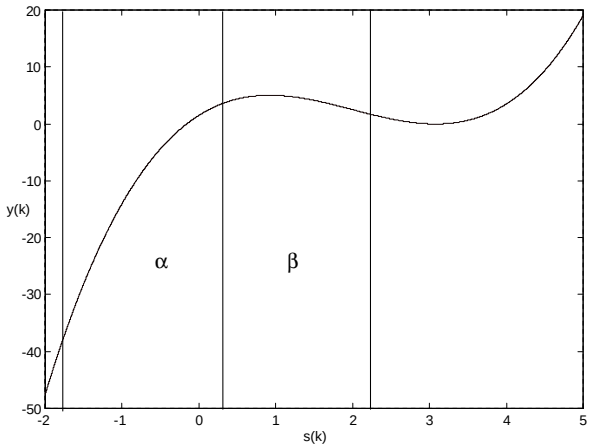


Figure 7: Interpolation and Extrapolation Domains

appreciated in Fig. 9 a-b. The indicator flag is in a tight correspondence with the error plot. It is important to note that the input belonging to the β *section* is $u(k) \in [.2, .5], k : 100, ..., 200$. However, the model is outside the generalization domain for $k : 100, ..., 225$. The evolution of the model outside the generalization domain for $k : 200, ..., 225$ is due to its own dynamic evolution and can not be explained directly for the input domain. However, the indicator flag points out exactly this situation. It is important to remark that the flag indicator highlight that the model is driven out the generalization domain, before it could be appreciated from the error plot.

3.4 Polytopic Domain

Let α_1 be the training data set, such that $\alpha_1 \subseteq \alpha$ the generalization domain of the LNet model. This set has $cardinal(\alpha_1) = 1000$ points and evaluating the inclusion of a new test point $\mathbf{x}$ should be done by checking if $\mathbf{x} \in \Im$ where $X = \alpha_1$. At this moment we want to reduce the polytope to a closed expression of the shell. For this we apply the algorithm described in section (2.3.1) and we arrive at a new set $\tilde{\alpha}_1$ with $cardinal(\tilde{\alpha}_1) = 82$ points. The results of each iteration of the shell evaluation algorithm are sumarized in table 1. In Fig. 10, the whole training set and the shell points are sketched, the"$\times$" marks are the shell points.

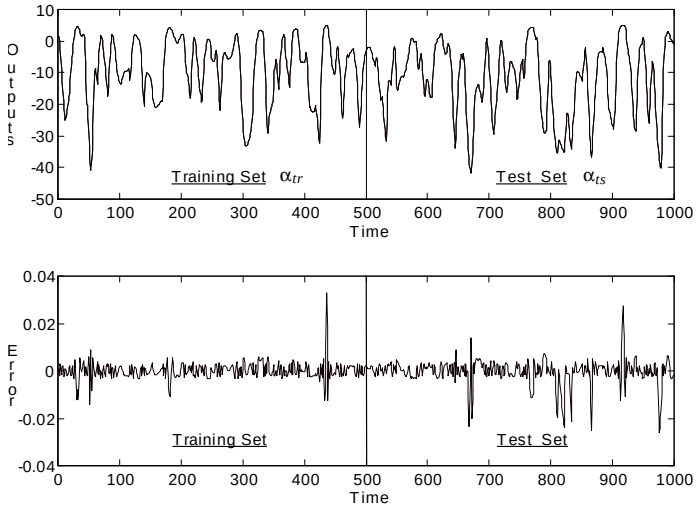


Figure 8: a) Model and Ssystem Outputs in sets α_{tr} and α_{ts}. b) Absolute Approximation Error.

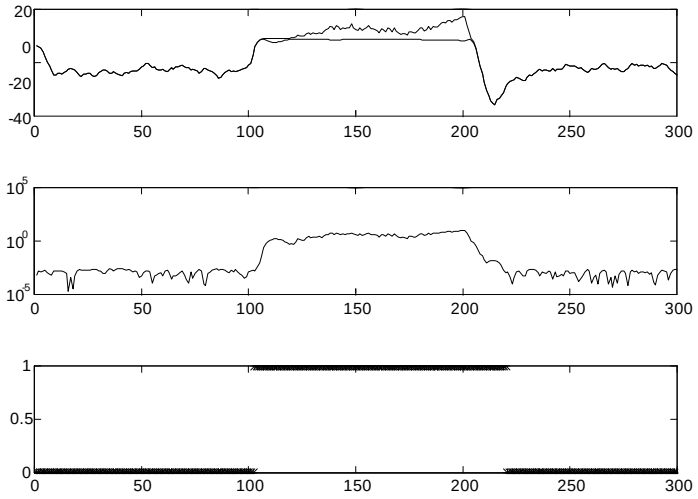


Figure 9: New signal exitation and validation checking

Table 1: Total Vertix per iteration

Iteration	Number of Elements
0	1000
1	116
2	88
3	82

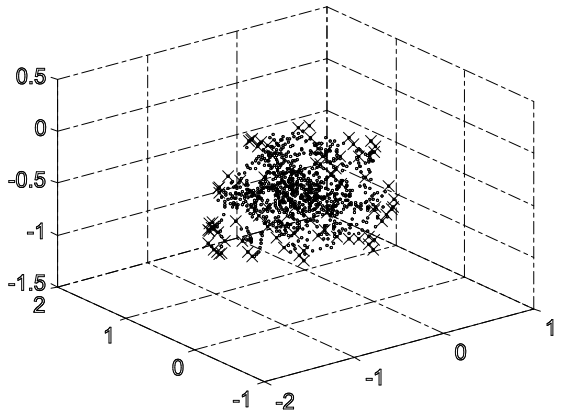


Figure 10: Generalization Domain and Shell

3.5 Domain Subscription

For this example the space dimension is $n = 3$. Now as shown in section (2.3.1) the number of planes needed to construct the parallelepiped is $2n = 6$. Making the principal components and finding the signed distances we arrived at the following planes:

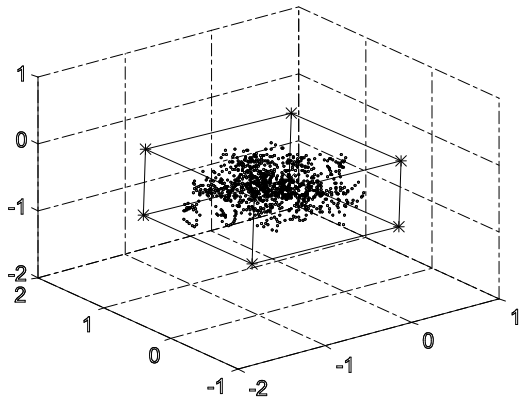


Figure 11: Domain Subscription

$$
\begin{array}{l}
-0.4198x_1 + 0.7468x_2 - 0.5158x_3 \geq 0 \\
-0.7335x_1 + 0.0555x_1 + 0.6774x_1 + 0.8186 \geq 0 \\
0.5345x_1 + 0.6627x_1 + 0.5245x_1 + 0.6591 \geq 0 \\
0.4198x_1 - 0.7468x_1 + 0.5158x_1 + 1.6990 \\
0.7335x_1 - 0.0555x_1 - 0.6774x_1 + 0.8035 \geq 0 \\
-0.5345x_1 - 0.6627x_1 - 0.5245x_1 + 0.3146 \geq 0.
\end{array}
,
$$

where all the director vectors point to the interior of the parallelepiped. Those faces are shown with the training data in Fig. 11.

4 Conclusions

The concepts of generalization and domain of generalization and their implications in modeling was addressed. We focused our attention on the generalization domain and its deterministic evaluation. This domain is usually related with the polytope formed by the data set. Briefly speaking, a given input is in the generalization domain if it is inner to that polytope. Three different approaches to the problem of determining if a given point is inside or outside of a given convex polytope were proposed. The first approach is based on the projection of the polytope to a reduced order space. If a given

point is inside the polytope, this condition is also guarantee for the reduced order polytope. The projected polytope was evaluated by using a finite algorithm. The second approach is based on an optimization approach. The formulations as a Linear Programming Problem as well as a Linear Matrix Inequality problem were considered. The last approach was focused on the simplification of the polytope definition. This was performed from two points of view. An algorithm to remove points from the interior of the polytope was presented. Also a Principal Component based methodology was discussed to define a simple hiperparalelepited that contains the original one. An example concerning with dynamic modeling was presented to show the availability of the whole approach.

References

[1] Agamennoni Osvaldo, Mandolesi Pablo and Desages Alfredo."On the evaluation of the generalization domain". *Proceedings of the World Conference on Neural Networks*. Washington D.C., pp II-70 - II-73. 1995.

[2] Boyd S. y L. Chua, "Fading Memory and the Problem of Approximating Nonlinear Operators with Volterra Series", *IEEE Trans. on Circuits and Systems*, **32**, 11, 1150-1161 (1985).

[3] Boyd Stephen, El Ghaoui Laurent, Feron Eric, and Balakrishnan V."*Linear Matrix Inequalities in System and Control Theory*". SIAM Studies in Applied Mathematics 1994.

[4] Courrieu Pierre."Three Algorithms for Estimating the Domain of Validity of a Feedforward Neural Network". *Neural Network, Vol. 7, No. 1*, pp. 169-174, 1994.

[5] Chvátal Vašek."*Linear Programming*". W. H. Freeman and Company. New York, 1983.

[6] Druker Harris and Le Cun Yann."Improving Generalization Performamce Using Double Backpropagation". *IEEE Transactions on Neural Network, Vol. 3. No. 6,pp. 991-997, Nov. 1992.*

[7] Holden B.S. and P.J.W. Rayner,"Generalization and PAC Learning: Some New Results for the Class of Generalized Single-Layer Networks,"

IEEE Transactions on Neural Networks, Vol. 6, No. 2, pp. 368-380, March 1995.

[8] Jayadeva and B. Bahumik,"A Neural Network for Determining the Planar Convex Hull", *Neural Network World, Vol. 2, No., 1, pp. 35-46,* 1992.

[9] Luenberger David."*Linear and Nonlinear Programming*" . Addison-Wesley Publishing Company.1984.

[10] Matthews M. B. y G. S. Moschytz, "The Identification of Nonlinear Discrete-Time Fading-Memory Systems Using Neural Network Models", *IEEE Trans. on Circuits and Systems II*, **41**, 11, 740-903 (1994).

[11] Nebot E., Sentoni G. and Masson F., Identification of a Flexible Manipulator Using Neural Networks, *IFAC'93, Sydney, Australia, July of 1993.*

[12] Preparata F. P. and M. I. Shamos (Eds.):"*Computational Geometry*", Springer Verlag, New York, 1988.

[13] Rockafellar R. T."*Convex Analysis*". Princeton University Press. 1970.

[14] Reed R., Marks R. J. and Oh S.. Similarities of Eerror Regularization, Sigmoid Gain Scaling, Target Smoothing and Training with Jitter. *IEEE Transactions on Neural Networks, Vol. 6, No. 3,pp 529-538, May 1995.*

[15] Sandberg Irwin W., "Approximation Theory for Discrete-Time Systems", *IEEE Trans. on Circuits and Systems*, **38**, 5, 564-566 (1991).

[16] Sentoni G., Agamennoni O. y J. Romagnoli, "On the Evaluation of a Certain Kind of Approximation Structure", *Latin American Applied Research*, **25**s, 35-40 (1995).

[17] Sentoni G., Agamennoni O., Desages A. y J. Romagnoli, "Approximate Models for Nonlinear Process Control", *AIChE Journal*, **42**, 8, 2240-2250 (1996).

[18] Vanderberghe Lien and Boyd Stephen."Semidefinite Programming" *Submited to SIAM Review* November 1994.

[19] Wahlberg Bo., System Identification Using Laguerre Models, *IEEE Trans. on Automatic Control Vol. 36, No. 5, pp. 551-561.,* May 1991.

[20] Webb A. R.,"Functional Approximation by Feed-Forward Networks: A Least Square approach to Generalization", *IEEE Transactions on Neural Networks, Vol. 5, No. 3, May 1994.*

[21] Xiangdong He. and Asada Haruhiko, A New Method for Identifying Orders of Input- Outputs Models For Nonlinear Dynamic Systems, *Proc. ACC'93, San Francisco, California., June 1993,* pp.2520–2523.

[22] Zervos C., Belanger P. R. and Dumont G.A., On PID controller tunin using orthonormal series identification, *Automatica, Vol 24, 165-175, 1988.*

[23] @@@Osvaldo Falta esta cita de la introduccion